Week 7

Research Paper Analysis, Presentation, and Practical Federated Learning

From Research Problem to Implementation and Evidence

Graduate course · English instruction



Faranak Solat
Department of AI & Data Science
Sejong University

What This Session Is For

Do not summarize a paper section by section. Reconstruct the research argument and test whether the evidence supports it.

Problem

What exact research problem is being addressed?

Method

What is the proposed mechanism, and why should it work?

Evidence

Do the experiments or analysis support the claim?

Today we will also build one small research story ourselves—from a failure mode to a coded solution.

What Every Presentation Should Cover

Motivation

Why does this problem matter?

Problem formulation

Clients, data, objective, assumptions, constraints.

Method

Core algorithmic idea—not every implementation detail.

Baselines

What alternatives are used for comparison?

Results

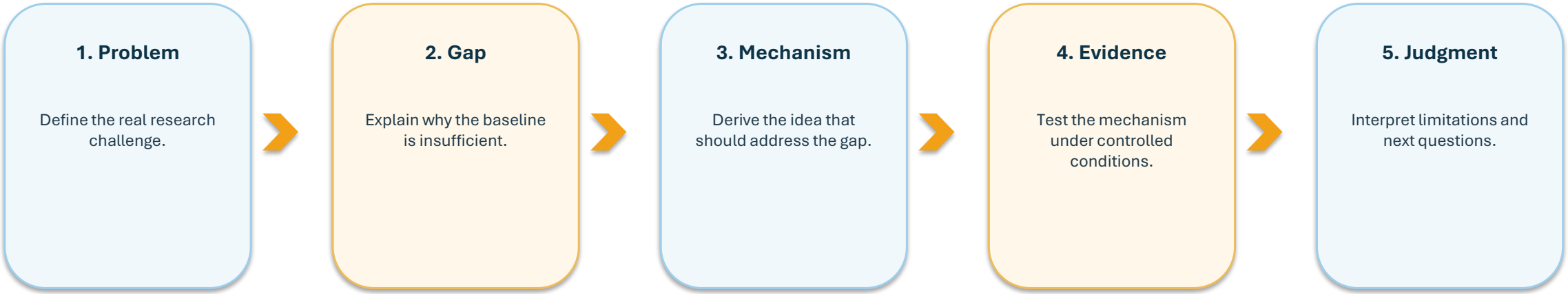
What do the experiments actually demonstrate?

Limitations

When might the method fail or become impractical?

The presentation should explain the logic of the paper—not reproduce its table of contents.

Recommended Research Story



The method should look like a logical response to the problem—not an arbitrary algorithmic trick.

How to Explain an Algorithm

1. Start with inputs / outputs

What does each client and the server know?

2. Identify the failure mode

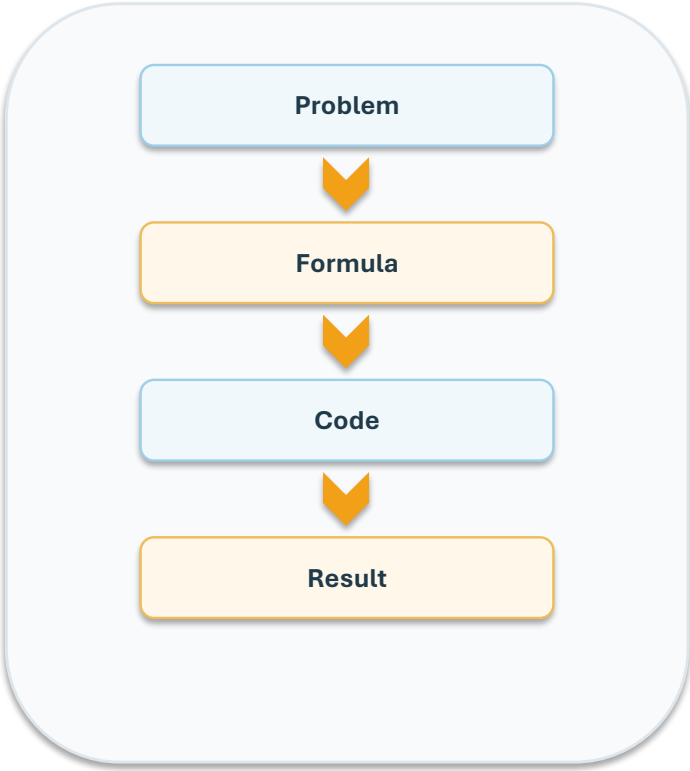
What goes wrong in the baseline?

3. Derive the modification

Which mathematical term directly targets that failure?

4. Translate to code

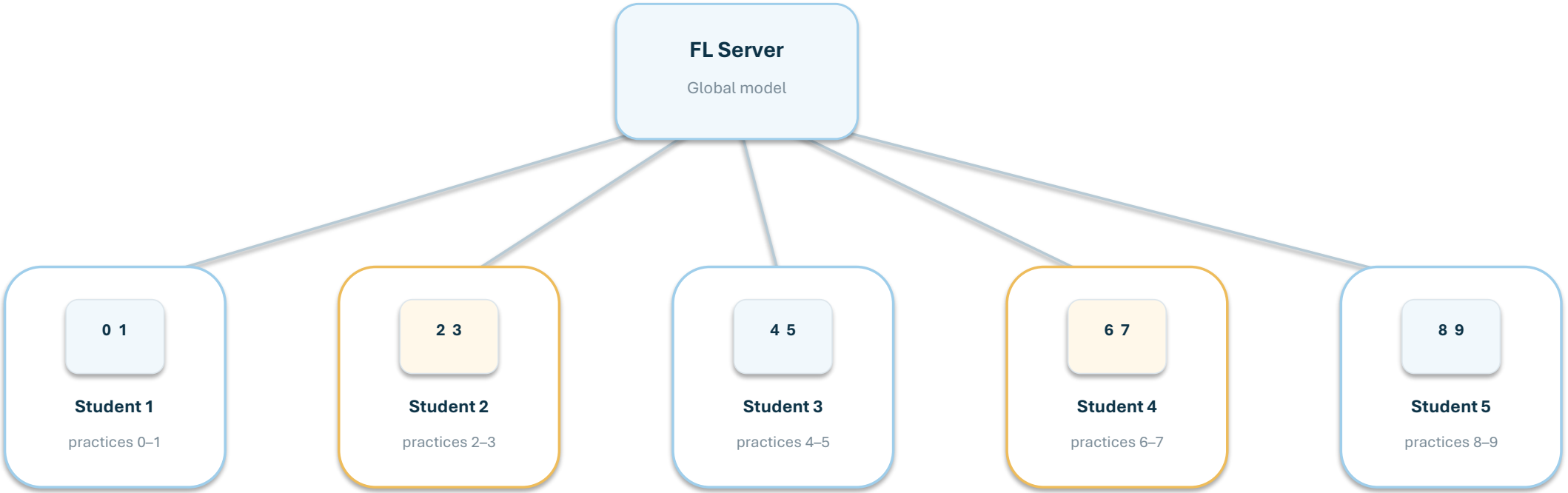
Show where the new mechanism enters the training loop.



We will now use FedAvg → FedProx as a worked example of this exact research flow.

Scenario: Federated Handwriting Recognition on Mobile Devices

A mobile learning app recognizes handwritten digits, but students' handwriting images must stay on their own phones or tablets.



Privacy constraint: raw handwriting samples remain local; only model information is exchanged.

The Real-World Complication: Users and Devices Are Different

The app has two simultaneous sources of heterogeneity.

Different local data

Students are working on different exercises, so each device sees a different label distribution.

Different device capability

Battery, CPU speed, and available training time differ, so devices cannot perform the same amount of local work.

User	Main digits	Device condition	Local steps
1	0–1	low battery / weak device	1
2	2–3	limited compute	5
3	4–5	average device	10
4	6–7	fast device	20
5	8–9	high compute / charging	40

Same global model, but very different local data + very different amounts of local training.

Question: what happens if a powerful device trains much longer on a highly biased local dataset?

From the Scenario to the Technical Problem

In our handwriting app, non-IID user data and unequal device capability cause local models to move by very different amounts before aggregation.

Statistical heterogeneity

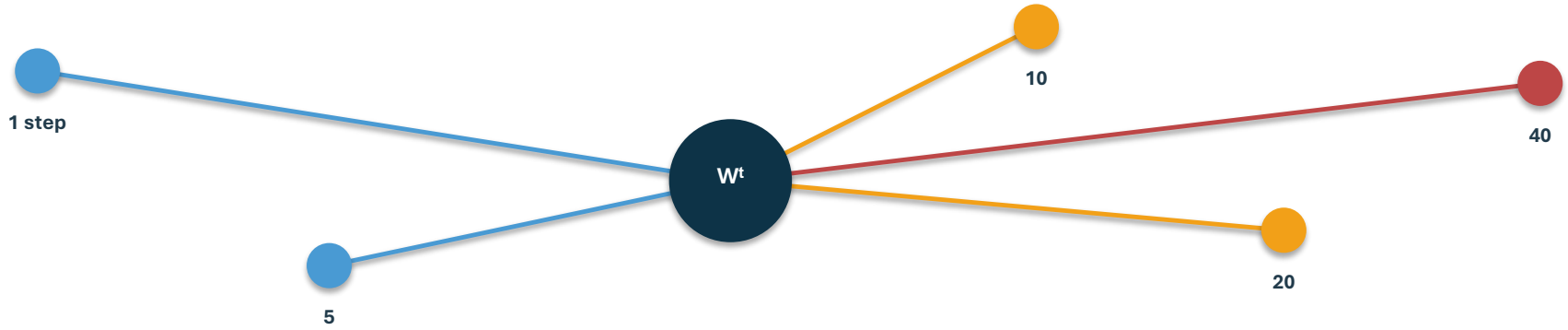
Each client mainly sees a different pair of digit classes.

System heterogeneity

Clients perform [1, 5, 10, 20, 40] local gradient steps.

Failure mode

Longer local training can amplify client drift under non-IID data.



Technical problem: preserve useful local learning while preventing excessive client drift.

Baseline: Why FedAvg Can Struggle

FedAvg aggregation: $W^{t+1} = \Sigma_k (n_k / n) W_k^{t+1}$

Local training

Client k minimizes only its local loss $F_k(W)$.

Under non-IID data

Different clients prefer different descent directions.

With unequal local work

Clients taking many more steps can move much farther from W^t.

Key limitation: averaging combines the final client models, but it does not explicitly control how far each client moves during local optimization.

So the failure mode originates inside local optimization—not only at the server.

Proposed Fix: Add a Proximal Constraint

FedProx modifies the local objective so that local adaptation is balanced against distance from the current global model.

$$\min_W \; F_k(W) + (\mu / 2) \|W - W^t\|^2$$

$F_k(W)$

Fit the client's own local data.

$\|W - W^t\|^2$

Penalize moving too far from the current global model.

μ

Controls the strength of the correction.

Gradient added by the proximal term: $\mu (W - W^t)$

Source: T. Li et al., "Federated Optimization in Heterogeneous Networks," MLSys 2020.

From the Formula to One Line of Code

The new mathematical term appears directly in the client-side gradient update.

Client-side update

```
grad = ce_gradient(W, X_local, y_local)

# proximal correction
if mu > 0:
    grad = grad + mu * (W - W_global)

W = W - lr * grad
```

FedAvg

Set $\mu = 0$ → ordinary local training.

FedProx

Set $\mu > 0$ → add a restoring term toward W^t.

Problem → formula → code: the implementation should visibly correspond to the proposed mechanism.

LIVE CODING IN COLAB

Implement the baseline, add the proximal term, and run the controlled comparison

1. Create heterogeneous clients
2. Implement FedAvg baseline
3. Add $\mu(W - W^t)$
4. Run FedAvg vs. FedProx

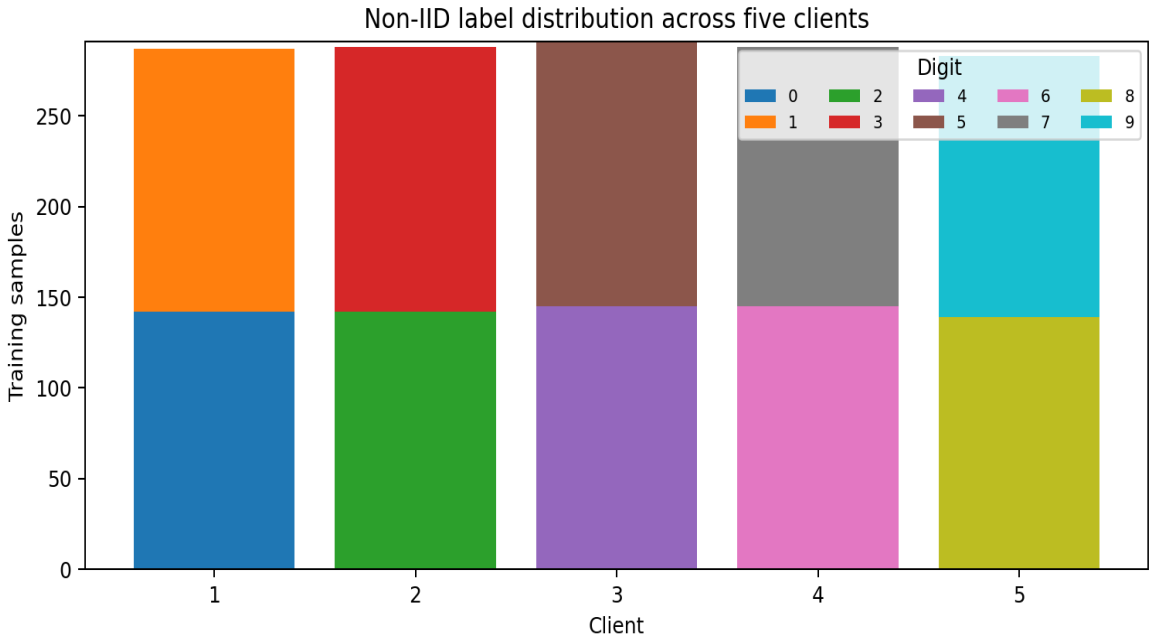
```
mu = 0.2

grad = ce_gradient(...)
grad += mu * (W - W_global)

FedAvg  : mu = 0
FedProx : mu = 0.2
```

Notebook: Week07_FedProx_Research_MiniLab.ipynb

Experimental Setup: Keep the Comparison Controlled



Dataset

Digits · 10 classes · same train/test split for both methods.

Clients

5 non-IID clients; each mainly owns two digit classes.

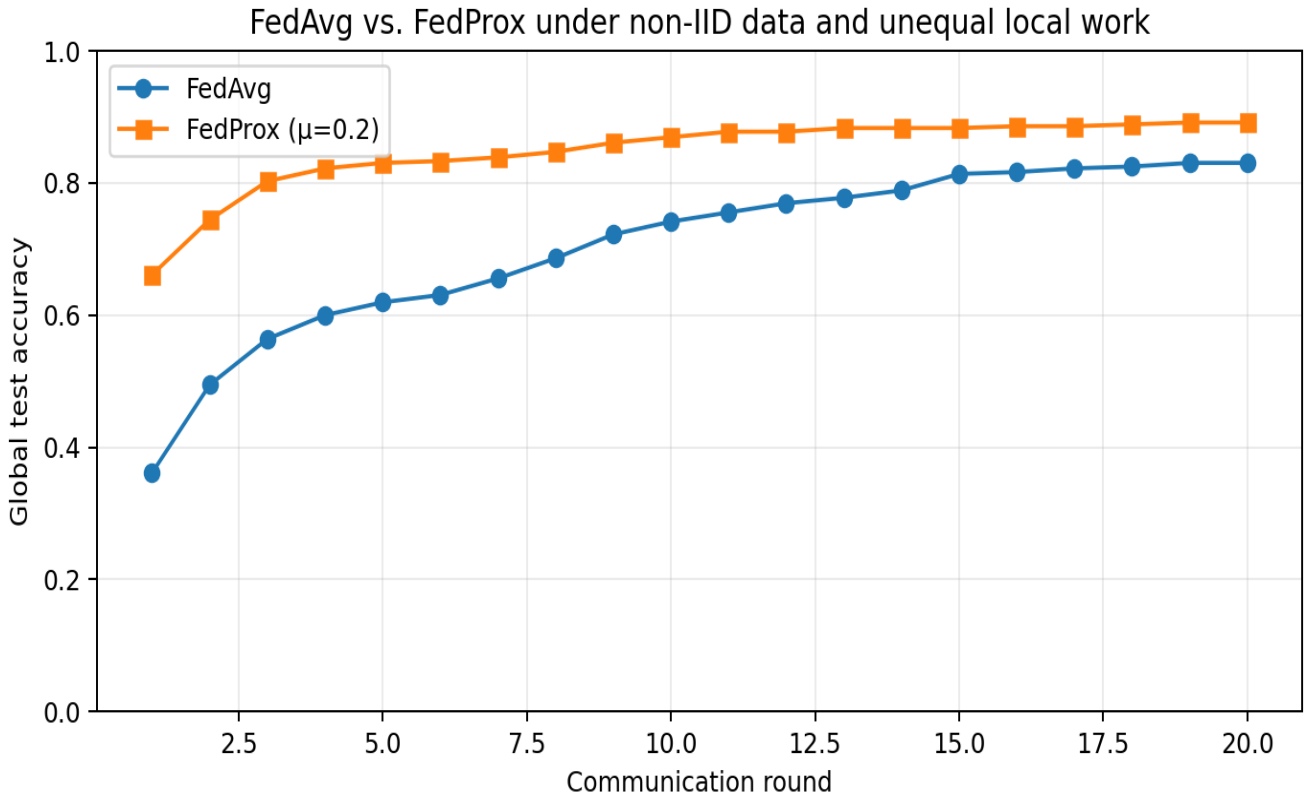
Local work

Steps = [1, 5, 10, 20, 40] to create system heterogeneity.

Same rounds, learning rate, data split, and aggregation rule; only the proximal correction changes.

FedAvg: μ = 0 | FedProx: μ = 0.2

Result 1: Did the Fix Improve Global Performance?



FedAvg

Final accuracy ≈ 0.831

FedProx

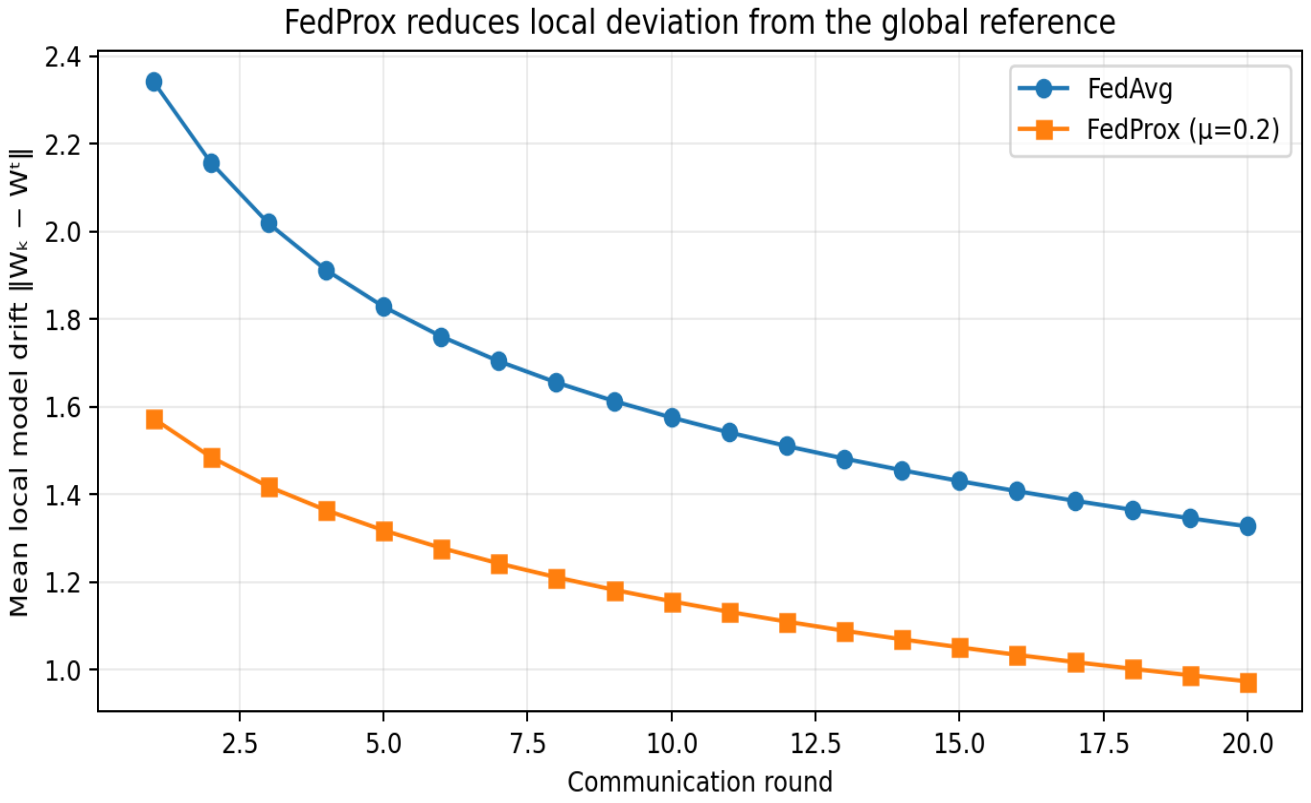
Final accuracy ≈ 0.892

Interpretation

In this setup, constraining local deviation improves the global result.

Accuracy gain in this fixed run: about +6.1 percentage points.

Result 2: Did the Mechanism Actually Reduce Client Drift?



FedAvg

Mean local drift ≈ 1.641

FedProx

Mean local drift ≈ 1.185

Mechanism check

The quantity targeted by the proximal term moves in the intended direction.

This is stronger evidence than accuracy alone: mechanism and outcome are consistent.

From a Classroom Demo to a Research Experiment

Hypothesis

The proximal term should help more when local drift is stronger.

Ablation

Sweep μ and study the trade-off between local freedom and global coupling.

Stress test

Increase non-IID severity or local-step imbalance.

Scientific reporting

Repeat with multiple seeds and report mean ± standard deviation.

Research flow: problem → hypothesis → mathematical mechanism → implementation → controlled comparison → interpretation.

How to Read Experimental Results

Dataset partition

Is the federated split realistic or synthetic?

Client participation

How many clients exist and participate per round?

Budget

Are rounds, local steps, model size, and communication comparable?

Metrics

Are client-level results reported or only a global average?

Baselines

Are strong and relevant methods included?

Variance

Are multiple seeds or confidence intervals shown?

Do not ask only whether the proposed method wins. Ask whether the comparison is fair.

Critical Reading Checklist

Assumptions

Which assumptions are necessary for the method?

Scalability

Will it still work with 10× or 100× more clients?

Privacy

Does the method expose additional client information?

System cost

What extra computation, memory, or messages are required?

Reproducibility

Are hyperparameters and protocols described clearly?

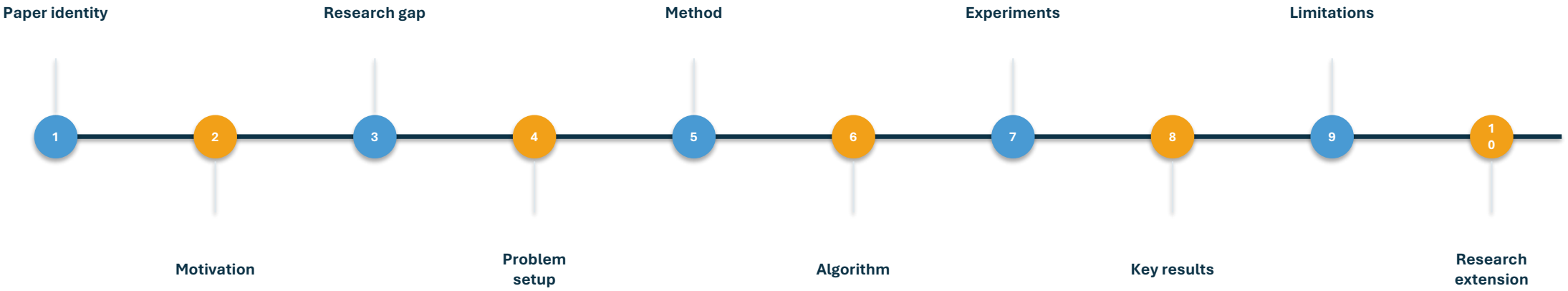
Claim strength

Does the paper claim more than the results justify?

Critical reading means understanding both what the paper proves and what it does not prove.

A Strong 10-Slide Student Presentation

One possible structure



The exact slide count can vary; the important thing is that the story remains coherent.

Suitable Paper Themes from Weeks 1–6

Federated optimization

FedAvg variants, convergence, local steps, optimization under heterogeneity.

Non-IID learning

FedProx, SCAFFOLD, FedNova, heterogeneity-aware optimization.

Systems

Stragglers, client selection, communication efficiency, resource-aware FL.

Personalization

Fine-tuning, FedPer, pFedMe, clustering, meta-learning.

Choose a paper with a clear technical contribution—not only a broad survey.

Peer Review During Presentations

One strength

Identify the most convincing part of the work.

One concern

Point to an assumption, experiment, or claim that needs scrutiny.

One question

Ask something that reveals deeper understanding—not a factual detail.

One extension

Suggest a feasible follow-up research direction.

Good peer review is specific, evidence-based, and constructive.

Presentation Standard

A graduate-level presentation should make the audience understand the research problem and be able to debate the method—even if they have not read the paper.

Clarity

Define new ideas clearly and explain why they matter.

Evidence

Show the strongest results, not every table in the paper.

Judgment

Explain what is important in the paper and what remains weak.

Do not confuse more slides with better explanation.

After Each Presentation

We will use the remaining time for structured discussion.

2–3 min

Clarifying questions

Facts, notation, setup

5–7 min

Critical discussion

Assumptions, evaluation, limitations

1 min

Takeaway

What research idea should we remember?

Instructor

Feedback

Technical accuracy, depth, communication, judgment

The discussion is part of the learning activity—not an interruption after the presentation.

Key Takeaways

1. **Present a paper as a research argument, not as a sequence of sections.**
2. A technical method should follow logically from a clearly stated failure mode.
3. **The formula, code, and experiment should all implement and test the same mechanism.**
4. Strong evidence checks both outcome metrics and the mechanism the method claims to improve.
5. Read like a reviewer. Present like a researcher.

Problem → mechanism → implementation → evidence → judgment

Q & A

Thank You.

sofar@sejong.ac.kr